Analytical Chemistry Cumulative Exam
September 29, 2023
Topic: Statistics and Error Propagation

You must show your working; no credit will be given for answers without working. Partial credit will be given for each step of the solution – A correctly set up problem calculator errors can still pass this cume!

Statistical tables and formulae are available at the end of the exam.

You should use the appropriate number of significant figures in all your answers.

Question 1 The concentration of F^- in solution (in mg/L) is related to the potential (*E*) of an ion-selective electrode by Conc. F^- (mg/L) = $10^{(K-E)/59\ mV}$ where *K* is a constant that depends on the electrode properties.

Your electrode has a calibration coefficient of *K* = 90±4 mV. You use it to measure a tap-water solution and get potential of *E* = 99±2 mV. What is the concentration of fluoride in the solution and what's the absolute error in this value? (use error propagation).

Answer format: Your answer should be in the form 12.345±0.6789 or 1.23×10^4±5.67 (where you should choose appropriate significant figures for both the answer and the error and you should choose appropriate units). *You are strongly encouraged to write all the steps of your calculation.*

Question 2

Two analytical methods are developed to measure the amount of lead in drinking water. Three samples of drinking water from different homes are analyzed using two analytical methods. The results of the two methods are below. Use an appropriate statistical test to determine if the methods differ significantly in the concentration of lead determined (confidence level 95%).

	Method 1 (Pb / ppb)	Method 2 (Pb / ppb)
Sample 1	1.21	1.93
Sample 2	12.2	12.7
Sample 3	0.71	1.12

Answer format: You answer should make it clear what method you are using for the statistical test, your calculated test statistic, the value you are using from the statistical tables, and the conclusions of your calculations (Either *Yes there is a significant difference at the 95% confidence level* **or** *No there is no significant difference at the 95% confidence level*).

Note: there is no known safe level for lead in drinking water, but and action level is set by the EPA at 15 ppb

Constants, Equations, and Tables

$$\bar{x} = \frac{\sum_{i=1}^{N} x_i}{N}$$

$$E = x_i - x_t$$

$$E_r = \frac{x_i - x_t}{x_t} \times 100\%$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}}$$

$$s = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \bar{x})^2}{N-1}} = \sqrt{\frac{\sum_{i=1}^{N} x_i^2 - \frac{\left(\sum_{i=1}^{N} x_i\right)^2}{N}}{N-1}}$$

$$s_m = \frac{s}{\sqrt{N}}$$

$$s_{pooled} = \sqrt{\frac{\sum_{i=1}^{N_1} (x_i - \overline{x_1})^2 + \sum_{j=1}^{N_2} (x_j - \overline{x_2})^2 + \sum_{k=1}^{N_3} (x_k - \overline{x_3})^2 + \ldots}{N_1 + N_2 + N_3 + \ldots - N_t}}$$

$$s_{pooled} = \sqrt{\frac{(N_1 - 1)\, s_1^2 + (N_2 - 1)\, s_2^2 + (N_3 - 1)\, s_3^2 + \ldots}{N_1 + N_2 + N_3 + \ldots - N_t}}$$

$$y = \frac{e^{-(x-\mu)^2/(2\sigma^2)}}{\sigma\sqrt{2\pi}} = \frac{e^{-z^2/2}}{\sigma\sqrt{2\pi}}$$

$$z = \frac{(x - \mu)}{\sigma}$$

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$$\mu = \bar{x} \pm \frac{z\,\sigma}{\sqrt{N}}$$

$$\mu = \bar{x} \pm \frac{t\,s}{\sqrt{N}}$$

$$z = \left| \frac{\bar{x} - \mu_o}{\sigma / \sqrt{N}} \right|$$

$$t = \left| \frac{\bar{x} - \mu_o}{s / \sqrt{N}} \right|$$

$$t = \left| \frac{\bar{x}_1 - \bar{x}_2}{s_{pooled} \sqrt{\frac{N_1 + N_2}{N_1 N_2}}} \right|$$

$$z = \left| \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{N_1 + N_2}{N_1 N_2}}} \right|$$

$$s_d = \sqrt{\frac{\sum \left(d_i - \bar{d}\right)^2}{N - 1}}$$

$$t = \frac{|\bar{d}|}{s_d / \sqrt{N}}$$

$$F = \frac{s_1^2}{s_2^2}$$

$$G_{calc} = \frac{|\text{questionable value} - \bar{x}|}{s}$$

$$RSD = \frac{s}{\bar{x}}$$

$$CV = \frac{s}{\bar{x}} \times 100\%$$

$$w = \left(x_{largest} - x_{smallest}\right)$$

	Function	Uncertainty
+/-	$y = x_1(\pm e_{x_1}) + x_2(\pm e_{x_2}) - x_3(\pm e_{x_3})$	$e_y = \sqrt{e_{x_1}^2 + e_{x_2}^2 + e_{x_3}^2}$
×/÷	$y = \dfrac{x_1(\pm e_{x_1}) \times x_2(\pm e_{x_2})}{x_3(\pm e_{x_3})}$	$\dfrac{e_y}{y} = \sqrt{\left(\dfrac{e_{x_1}}{x_1}\right)^2 + \left(\dfrac{e_{x_2}}{x_2}\right)^2 + \left(\dfrac{e_{x_3}}{x_3}\right)^2}$
	where $\%e_{x_1} = 100\% \dfrac{e_{x_1}}{x_1}, \%e_{x_2} = 100\% \dfrac{e_{x_2}}{x_2}$, $\%e_{x_3} = 100\% \dfrac{e_{x_3}}{x_3}, \%e_y = 100\% \dfrac{e_y}{y}$	or $\%e_y = \sqrt{(\%e_{x_1})^2 + (\%e_{x_2})^2 + (\%e_{x_3})^2}$
exponentiation	$y = x(\pm e_x)^a$	$\dfrac{e_y}{y} = a\,\dfrac{e_x}{x}$
		or $\%e_y = a\ \%e_x$ where $\%e_y = 100\% \dfrac{e_y}{y}, \%e_x = 100\% \dfrac{e_x}{x}$
logarithm	$y = \log_{10}(x(\pm e_x))$	$e_y = \dfrac{1}{\ln 10}\dfrac{e_x}{x} = 0.43429\ \dfrac{e_x}{x}$
	$y = \ln\left(x(\pm e_x)\right)$	$e_y = \dfrac{e_x}{x}$
antilogarithm	$y = \text{antilog}_{10}(x(\pm e_x)) = 10^{x(\pm e_x)}$	$\dfrac{e_y}{y} = (\ln 10)\, e_x \approx 2.3026\, e_x$

TABLE 4-5 Critical values of *G* for rejection of outlier

Number of observations	*G* (95% confidence)
4	1.463
5	1.672
6	1.822
7	1.938
8	2.032
9	2.110
10	2.176
11	2.234
12	2.285
15	2.409
20	2.557

$G_{calculated}$ = |*questionable value − mean*|/*s*. *If* $G_{calculated} > G_{table}$, *the value in question can be rejected with 95% confidence. Values in this table are for a one-tailed test, as recommended by ASTM.*

TABLE 4-4 Values of Student's *t*

Degrees of freedom	Confidence level (%) 50	90	95	98	99	99.5	99.9
1	1.000	6.314	12.706	31.821	63.656	127.321	636.578
2	0.816	2.920	4.303	6.965	9.925	14.089	31.598
3	0.765	2.353	3.182	4.541	5.841	7.453	12.924
4	0.741	2.132	2.776	3.747	4.604	5.598	8.610
5	0.727	2.015	2.571	3.365	4.032	4.773	6.869
6	0.718	1.943	2.447	3.143	3.707	4.317	5.959
7	0.711	1.895	2.365	2.998	3.500	4.029	5.408
8	0.706	1.860	2.306	2.896	3.355	3.832	5.041
9	0.703	1.833	2.262	2.821	3.250	3.690	4.781
10	0.700	1.812	2.228	2.764	3.169	3.581	4.587
15	0.691	1.753	2.131	2.602	2.947	3.252	4.073
20	0.687	1.725	2.086	2.528	2.845	3.153	3.850
25	0.684	1.708	2.060	2.485	2.787	3.078	3.725
30	0.683	1.697	2.042	2.457	2.750	3.030	3.646
40	0.681	1.684	2.021	2.423	2.704	2.971	3.551
60	0.679	1.671	2.000	2.390	2.660	2.915	3.460
120	0.677	1.658	1.980	2.358	2.617	2.860	3.373
∞	0.674	1.645	1.960	2.326	2.576	2.807	3.291

In calculating confidence intervals, σ may be substituted for s in Equation 4-7 if you have a great deal of experience with a particular method and have therefore determined its "true" population standard deviation. If σ is used instead of s, the value of t to use in Equation 4-7 comes from the bottom row of this table.

Values of t in this table apply to two-tailed tests illustrated in Figure 4-9a. The 95% confidence level specifies the regions containing 2.5% of the area in each wing of the curve. For a one-tailed test, we use values of t listed for 90% confidence. Each wing outside of t for 90% confidence contains 5% of the area of the curve.

TABLE 4-4 Critical values of $F = s_1^2/s_2^2$ at 95% confidence level

Degrees of freedom for s_2	Degrees of freedom for s_1: 2	3	4	5	6	7	8	9	10	12	15	20	30	∞
2	19.0	19.2	19.2	19.3	19.3	19.4	19.4	19.4	19.4	19.4	19.4	19.4	19.5	19.5
3	9.55	9.28	9.12	9.01	8.94	8.89	8.84	8.81	8.79	8.74	8.70	8.66	8.62	8.53
4	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.75	5.63
5	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.50	4.36
6	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.81	3.67
7	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.58	3.51	3.44	3.38	3.23
8	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.08	2.93
9	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.86	2.71
10	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.84	2.77	2.70	2.54
11	3.98	3.59	3.36	3.20	3.10	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.57	2.40
12	3.88	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.47	2.30
13	3.81	3.41	3.18	3.02	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.38	2.21
14	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.31	2.13
15	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.25	2.07
16	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.19	2.01
17	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.15	1.96
18	3.56	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.11	1.92
19	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.07	1.88
20	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.04	1.84
30	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.84	1.62
∞	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.46	1.00

Critical values of F for a one-tailed test of the hypothesis that $s_1 > s_2$. There is a 5% probability of observing F above the tabulated value.

You can compute F for a chosen level of confidence with the Excel function FINV(probability,deg_freedom1,deg_freedom2). The statement "=FINV(0.05,7,6)" reproduces the value F = 4.21 in this table. The statement "=FINV(0.1,7,6)" gives F = 3.01 for 90% confidence.

Harris, *Quantitative Chemical Analysis*, 8e
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